THE WORLD IN SPACE — A Home That's Falling Around the Earth (7.7, Part 2 of 2)

Student: Mark | **Duration:** ~60–65 min | **Text:** Spectrum Science G3, Lesson 7.7, "The World in Space" (pp. 116–117) | **Role: "Flight Director Mark"** (carried over from Part 1) — today he moves into the house that the race built.

The narrative spine: Part 1 ended with two rivals shaking hands in orbit. Today Flight Director Mark moves into the house that handshake started building: he watches the live feed from its cameras, goes inside it, learns to find it in the sky over Xi'an — and then faces the mystery planted last lesson: the ISS is falling the whole time, so why is everything inside floating? The answer (orbit = free fall, gravity is ~90% as strong up there) is the science core, cracked with a Hadfield video and a PhET simulation. The text is read as the station's "official history file," the lesson tours the stations that came before the ISS, and it ends with Mark explaining one sentence in his own words.

The takeaway sentence: "There IS gravity on the space station — the station and everything in it are falling around the Earth together, and that's why everything floats."

Resources

- [VIDEO] **ISS live Earth camera** (real-time external feed; black when the station is on Earth's night side — that's a teachable moment): isslivenow.com (aggregator of NASA's official feed)
- [HTML] **ESA ISS panoramic tour** (360° interactive, inside the real modules): esa.int/.../International_Space_Station_panoramic_tour (NASA's hub for it: nasa.gov/feature/iss-virtual-tour)
- [HTML] **ISS Explorer** (Unity web app, 3D model + module map — the screenshots in `screenshots/` in this folder; great for the PORT/STBD/OVHD/DECK hunt and the module-name decoder)
- [VIDEO] **Chris Hadfield, "Tears in Space (Don't Fall)" (~2.5 min):** youtube.com/watch?v=P36xhtpw0Lg
- [VIDEO] **Chris Hadfield, "How to brush your teeth in space" (~3 min):** youtube.com/watch?v=3bCoGC532p8
- [VIDEO] **Chris Hadfield, "How astronauts sleep in space" (~3 min):** youtube.com/watch?v=UyFYgeE32f0
- ☒ **Heavens-Above** (ISS + Tiangong flyovers over Xi'an): heavens-above.com — set observing site to Xi'an before class (no account needed; NASA's Spot the Station moved sighting lookups into a mobile app in June 2025, this is the no-install alternative)
- [HTML] **PhET "Gravity and Orbits":** phet.colorado.edu/en/simulations/gravity-and-orbits
- [VIDEO] **Space stations history video (~5.5 min, Salyut → ISS → Tiangong → future):** youtube.com/watch?v=_CY9Z3XtSBs — used in Activity VII; transcript checked and historically accurate
- ☒☒ Station lineup pictures for Activity VII (same set as Part 1): **Salyut 1** (1971) — en.wikipedia.org/wiki/Salyut_1, **Skylab** (1973) — en.wikipedia.org/wiki/Skylab, **Mir** (1986) — en.wikipedia.org/wiki/Mir, **ISS** (first module 1998) —

en.wikipedia.org/wiki/International_Space_Station

- [VIDEO] Bonus: **more ISS experiments** (youtube.com/watch?v=IvdohUPWg-0) and more Hadfield clips (wringing out a wet towel — the water crawls up his arms, brushing teeth is already in Activity II)
- [VIDEO] Bonus: "weird experiments on the ISS" — growing plants, bacteria in space (search-free option: ESA/NASA YouTube channels)

ACTIVITY 0 — SPEED RECAP (5 min) • true/false sprint from Part 1

Format: rapid verbal round before anything new — ten statements, hands up true / down false, no pausing, answer + one-line why immediately after. It's a game, not a test.

1. "Humans had nothing in space before 1900." (**TRUE** — first was Sputnik, 1957)
2. "The first thing ever in space was American." (**FALSE** — the V-2 rocket, Germany, WWII)
3. "The USA got the first human into space." (**FALSE** — Gagarin, USSR, 1961)
4. "America's first satellite launch crashed on live TV." (**TRUE** — Vanguard, 1957, "Flopnik")
5. "The USSR got almost all the 'firsts.'" (**TRUE** — satellite, animal, human, spacewalk)
6. "Both sides lost astronauts during the race." (**TRUE** — Apollo 1, Soyuz 11, Challenger)
7. "The Saturn V is still the most powerful rocket ever flown." (**FALSE** — SpaceX Starship, about twice the thrust)
8. "The space race ended with the countries still enemies." (**FALSE** — Apollo–Soyuz handshake, 1975)
9. "The two rivals built the ISS together." (**TRUE** — first piece 1998)
10. "The ISS is falling right now." (**TRUE** — and that's why we're here today!)

Question 10 is the springboard: "Last time I told you the space station is falling around the Earth every single second. Today we move in and figure out what that's like. Flight Director, welcome aboard."

ACTIVITY I — LIVE FROM ORBIT (3 min) • real-time camera

Site: ISS live Earth camera (real-time external feed; black when the station is on Earth's night side — that's a teachable moment) — isslivenow.com (aggregator of NASA's official feed). Full screen, no talking for the first 20 seconds. Let the image do the work.

1. "This is not a recording. This is happening right now."
2. "Is it day or night up there at this moment? How can you tell?" (If black: it's over Earth's night side — the station circles the whole planet every 92 minutes, so it gets a sunrise/sunset every 45 minutes. Daylight comes back soon.)
3. One more mystery planted: "Look at the astronauts in the little corner views when they appear — which way is up for them?" (There is no up. Hold that thought.)

ACTIVITY II — MOVE IN: 360° TOUR (12 min) • scavenger hunt inside the real ISS

Sites: ESA ISS panoramic tour — esa.int/.../International_Space_Station_panoramic_tour — and/or **ISS Explorer** (the Unity web app from the screenshots/ folder). Screen-shared, teacher drives the mouse at HIS command.

"Welcome aboard. Here's your job: I've got five things for you to find. Tell me where to fly."

Scavenger-hunt list (adjust to what the tour shows): the toilet ☒, the gym (exercise bike/treadmill — crucial: without it astronauts' muscles waste away), the cupola (the station's "window with the best view in the universe" — seven windows facing Earth), a sleeping pod (they sleep strapped to the wall so they don't drift off), the kitchen/galley (food in pouches).

After each find: "Would you be able to live like this for six months? What's the one thing you'd miss most?" (Expected: showers, real food, walking. Personal connection — what about YOU.)

The ship-words hunt (~2 min — ISS Explorer is ideal, the labels are on the hatches): somewhere in the module he will find four labels: **PORT**, **STBD**, **OVHD**, **DECK**. Ask him to guess before revealing: "This is a machine, but it uses SAILING words. Which one is the floor?" → DECK = floor, OVHD = overhead = ceiling, STBD = starboard = RIGHT side, PORT = left side. Then the two fun questions: (1) "Why sailing words on a spaceship?" — because spacecraft are run like ships: early capsules were flown by military pilots and Navy men, and sailors had spent centuries agreeing on words for "left, right, floor, ceiling" — a spaceship is a vessel, so it inherited the whole vocabulary (bridge, deck, hatch, crew). (2) "Why do they even NEED the words?" — because in weightlessness there is no natural "up"! The station must agree on one direction and print it on the walls, or "left" and "floor" would mean different things depending on which way you were floating. Have him notice: on the screenshot, **DECK is printed upside down** — it's written so you can read it coming through the hatch. Fun fact for the word "starboard": old Viking ships had the steering oar on the RIGHT side ("steer-board"), so the ship always docked with its left side against the port — left = "port", right = "steer-board". A thousand-year-old habit, painted inside a space station in 1998.

The module names (~1 min — show the module map screenshot): the ISS rooms have real names, and half of them are ordinary words in other languages — a free vocabulary lesson:

- **Zvezda** = "Star" ☒ and **Zarya** = "Dawn" (Russian — Alex can pronounce these for him!), **Nauka** = "Science", **Prichal** = "Pier" (where the ships dock), **Rassvet** = "Morning", **Poisk** = "Search"
- **Kibo** = "Hope" (Japanese lab), **Columbus** (European lab — after the explorer)
- **Tranquility** — named after the Sea of Tranquility, the Apollo 11 landing site; **Unity** and **Harmony** — named by American schoolkids in national contests
- **Cupola** = "little dome" (Latin) — the seven-window viewing dome; **Quest** = the airlock

Everyday life (pick ONE, ~3 min): watch one Hadfield clip and mine it:

- **"How to brush your teeth in space"** — youtube.com/watch?v=3bCoGC532p8. Questions: Why is there no tap or sink? (No running water — water would float everywhere.) Where does the used toothpaste GO? (He SWALLOWS it — it's edible, and there's no sink to spit into!) Why doesn't toothpaste go up your nose? (Nothing pushes it — it just floats.)
- **"How astronauts sleep in space"** — youtube.com/watch?v=UyFYgeE32f0. Questions: Why no mattress and no pillow? (No gravity — nothing to hold you up; every muscle can relax, your arms float up in front of you.) Where does Hadfield say his sleep station is? ("It's actually on the

floor, but once you're inside, you just can't tell" — the labels decide where the floor is, remember?)

Escape hatch: the tour can be fiddly with dragging. If it fights you, switch to a pre-loaded gallery of interior photos and keep the same scavenger list — the objects are what matter.

ACTIVITY III — FIND IT IN THE SKY (7 min) • Heavens-Above challenge

Site: Heavens-Above — heavens-above.com (observing site preset to Xi'an).

"Right now, while we sit here, there are people flying over your head at 28,000 kilometers per hour. Your challenge, Flight Director: find the place on Earth from which you can see the ISS — and figure out when it flies over Xi'an this week."

1. **The map first:** show the current ground track / the pass list. The challenge wording matters — he's not just reading a table, he's answering "where on the planet do you have to stand to see it right now?" (Somewhere along that line — and if the line doesn't cross Xi'an, you wait for the next orbit.)
2. **Read the sky-map panel like an astronomer** (use the screenshot in screenshots/, or live in the app): **Altitude 39.6°** = how high above the horizon it appears (90° = straight overhead — the higher, the better the view). **Azimuth 120.6°** = compass direction to face (0° = north, 90° = east, 180° = south — so 120° is east-southeast). **Range 646 km** = how far away the station is at this moment — much more than its 400 km height because you're seeing it off toward the horizon. **Magnitude -3.1** = brightness — and here's the trick: the LOWER (more negative) the number, the BRIGHTER the object. -3 makes the ISS brighter than any star, about like Venus. **Orbit 415 × 425 km; 51.6°** = it circles Earth at a height between 415 and 425 km (the orbit is slightly oval) and the orbit is tilted 51.6°, which is why it can pass over almost everywhere people live — including Xi'an.
3. **Pick the best pass** (brightest/highest/longest) and screenshot it — he takes it home. Guess before revealing: "How high do you think it is? How fast?" → ~400 km up (roughly the distance from Xi'an to Luoyang, straight up), 28,000 km/h, one lap every ~92 minutes.
4. **Other satellites moment:** the same sky map shows dozens of **Starlinks** — whole strings of them. "See how crowded it's getting up there? All of those are satellites launched since the space race. The race built the road; now everyone drives on it." Heavens-Above also lists **Tiangong** — China's own station. "Your country has one too. Two stations — just like the race we studied⋯ except now everyone shares."

Escape hatch: if the site is having a bad day, show a screenshot of a pre-checked pass list instead — the magic is the flyover itself, not the live lookup.

ACTIVITY IV — THE MYSTERY: TEARS IN SPACE (6 min) • video

Video: "Tears in Space (Don't Fall)" (~2.5 min) — youtube.com/watch?v=P36xhtpw0Lg — Chris Hadfield, commander of the ISS.

Format: doing first, explanation after. Framing before play: "Watch what happens to water up there — and remember the station is FALLING while this happens."

Comprehension questions (verbal, quick):

1. Why did Hadfield use drinking water instead of real tears? (He can't cry on command.)
2. What happened to the water he put in his eye? (It stayed as a blob on his eye — didn't fall, didn't drip.)
3. What happens if you keep "crying" in space? (The ball of water grows until it crosses your nose and gets into your OTHER eye — or evaporates, or spreads over your cheek.)
4. Why didn't it fall? (Listen for: "no gravity" — DON'T confirm yet. Park it.)
5. What was his advice? ("Grab a hanky." ☒)

Park the mystery for the next activity: "So why does the water just sit there as a ball? Is it because there's no gravity? Everyone says that. Let's check with a machine that can measure gravity."

ACTIVITY V — PHET: MEASURING GRAVITY (9 min) • simulation

Sim: PhET "Gravity and Orbits" — phet.colorado.edu/en/simulations/gravity-and-orbits. How it actually works (per the teacher guide): you can drag objects to reposition them, change velocity by dragging the **v at the tip of the velocity arrow**, adjust mass with sliders (0.5×–2×), **turn gravity off entirely**, and switch between the "Model" screen (enlarged, easy to play) and the "To Scale" screen (real distances).

"Everyone says there's no gravity in space. Let's check. Open the Earth–Space Station system and watch the gravity arrow on the station — which way does it point, and how big is it?" (Points straight at Earth — gravity up there is ALMOST as strong as on the ground, ~90%.)

"So if gravity is pulling the station down⋯ why doesn't it come down?" (It's moving sideways FAST — 28,000 km/h. It's falling, but it keeps missing the ground — Newton's cannon from last lesson!)

Then let HIM break it, three rounds:

1. **Turn gravity OFF** (the button exists — predict first: "What happens the moment gravity disappears?") → the station flies away in a straight line. Gravity is the invisible string bending the path into a circle.
2. **Pause the sim ☒, drag the station farther from Earth, press play** → it orbits SLOWER on a longer path. Closer in → faster. (This is the correct way to move it — pause, drag, play.)
3. **Drag the tip of the velocity arrow shorter** (make the station slower) → watch the orbit shrink and spiral down. Make it longer → it swings out into a bigger orbit or escapes. "Speed is the only thing standing between the station and the ground."

- (Optional if he's eating it up: the To Scale screen for real distances; or double the station's mass with the slider and watch the orbit not change — heavy things fall just as fast.)

The reveal — make HIM say it: "So why do the tears blob instead of fall? You said it yourself before the video — is there really no gravity?" → Goal: the station, Hadfield, and the water are ALL falling together — everything is in free fall, so nothing pulls "down" relative to anything else. "'Zero gravity' is really 'everything falling together.'"

Escape hatch: if PhET is too finicky mid-call, the whole activity still works verbally: stand up, jump — "for a split second while you fell, you were weightless, exactly like an astronaut, all the time." But try the sim first.

ACTIVITY VI — TEXT: THE STATION'S OFFICIAL FILE (9 min) • reading as evidence

Format: the Spectrum page, screen-shared, teacher reads aloud, Mark follows. This is ⩽¼ of the lesson and must never feel like the main event. Frame it as the file on Flight Director's desk.

Tracking job, given before reading: "This is the station's official history file. Three things to hunt while I read: (1) every NUMBER on the page — write them down. (2) One sentence that tells you roughly when this page was written. (3) Anything that sounds out of date by now — this file is old."

- Pause for vocabulary as they come: **space station**, **launch**, **orbit** — Mark finds the sidebar definitions first, then rephrases in his own words.
- React to facts along the way, don't save reactions: 17,000 mph, 92 minutes, football-field size, six-month stays, wealthy tourists.
- The out-of-date catches (he may find some himself; nudge if not):
- "a one year mission is **being planned for 2015**" — that's over a decade ago; year-long missions have now been done (Kelly, 2015–16 — the page almost predicted it).
- Q4 "completed in the year 2000" is false — assembly finished in 2011, and the text itself says finished in 2010. (Let him catch the contradiction.)
- "Six or seven astronauts will be able to live there" — close enough, but usually there are 7, and for a while there were⋯ more.
- Workbook Q6–9 (skip 1–5, mostly covered by the lesson): Q6 why not built on Earth (too big to launch whole — built piece by piece in orbit); Q7 why countries worked together (too expensive alone; each builds what it's best at); Q8 why study Earth from space (see the big picture — hurricanes, pollution, whole forests); Q9 $100 billion well spent? — his opinion, verbal, no wrong answers, but he must back it with one fact from today.

Escape hatch: if reading energy drops, cut to Q6 and Q9 only — one evidence question, one opinion question.

ACTIVITY VII — STATIONS: FROM SALYUT TO THE ISS (10 min) • lineup game + history video

Video: space stations history (~5.5 min, Salyut 1 → Skylab → Salyut program → Mir → ISS → Tiangong → future stations) — youtube.com/watch?v=_CY9Z3XtSBs. Transcript checked: historically accurate.

Pictures (screen-share one at a time for the warm-up, NO captions — he guesses before you reveal): Salyut 1 (1971) — en.wikipedia.org/wiki/Salyut_1, **Skylab** (1973) — en.wikipedia.org/wiki/Skylab, **Mir** (1986) — en.wikipedia.org/wiki/Mir, **ISS** (first module 1998) — en.wikipedia.org/wiki/International_Space_Station. (This is the station-lineup game that used to live in Part 1's Activity V — it moved here, where stations are the whole point.)

Callback to open: "Last lesson ended with a handshake — two rivals docking in 1975. Know what they started building right after that? Houses. In space. Let's see if you can recognize the houses."

1. **Guessing game, fast** (~2 min): show the four pictures out of order, one at a time, no captions. For each: "Which country built this one? And when — before or after the Moon landing? Before or after the handshake?" He argues from the look (one-module vs grown-like-a-house vs huge-and-white). Reveal after each guess.
2. **The video** (~5.5 min) — then mine it with transcript-based questions (~2 min):

- "The first station ever — Salyut 1. What happened to its FIRST crew?" (Soyuz 10 couldn't dock at all — came home without getting in!) "And the second?" (Soyuz 11 — a record 23 days inside⋯ and the crew died on the way home. The same tragedy that's on our race timeline.)
- "America's Skylab was built from WHAT?" (A leftover piece of a Saturn V rocket — recycling! And it was damaged at launch — a solar panel and its shield ripped off — so the first crew had to fix it on a daring spacewalk.)
- "Mir survived WHAT?" (Fires, a collision, and the collapse of the Soviet Union itself — and Americans came to live on it: Shuttle–Mir.)
- "Since when has someone ALWAYS been living on the ISS?" (November 2000 — the station has been inhabited longer than Mark has been alive.)
- "What does Tiangong mean, and what number is China on the list of station-owning countries?" ("Heavenly Palace" — and THIRD, after the USSR/Russia and the USA.)
- "Where is the next station going to orbit, and who might run future ones?" (The Moon — the Lunar Gateway, for the Artemis missions; and companies like Axiom Space — commercial stations, space hotels!)

3. **Land the arc** (~30 sec): "The race ended. But the stations didn't stop — they got bigger, and the rivals built one together. It's still up there, with people inside, today. And the next ones are already being built — one of them around the Moon."

Xi'an note: today the story continues — Tiangong, China's station, is the newest chapter of the same book.

ACTIVITY VIII — MISSION DEBRIEF (5 min) • the one-sentence verdict

Format: the mission log, same ritual as Part 1.

"Flight Director, final log entry. Last time you told me how the space race ended. This time, the mystery: 'The space station has no gravity' — true or false, and WHY?" → Goal: "There's gravity up there — but the station and everything in it are falling around the Earth together, so everything floats." Help him compress; accept his words, then offer the tight version.

End on the fun note (science-on-board bonus, pick one if time):

- [BONUS] **More ISS experiments** — youtube.com/watch?v=IvdohUPWg-0 (2–3 min of floating experiments)
- [BONUS] Hadfield wringing out a wet towel in space (the water crawls up his arms instead of dripping — search "Hadfield wet towel" on YouTube before class; 3 min)
- [BONUS] Weird ISS experiments: plants grow sideways and confused without "down"; bacteria get stronger in space — scientists are still figuring out why
- [BONUS] Trivia flex: astronauts get taller in space (their spines stretch a few centimeters) — then shrink back on Earth

Final bridge: "Next time you look up at the right moment, you'll know there are people falling around the planet with you. And if you spot it over Xi'an — that's your homework. Not from me. From the sky."

Notes on Mark

- **Carry the role over:** he earned "Flight Director" last lesson — use it constantly. The arc rule: same mission, new phase (race → residence).
- **The mystery structure matters:** Activity V only lands because the question was planted twice — in Part 1 ("the ISS is falling") and here in Activity I ("which way is up?"). If he doesn't remember, remind him — recursion is fine, being told cold is not.
- **Verbal first, always:** the free-fall reveal must come out of HIS mouth. Resist explaining; ask "so what is pulling the water down?" and wait.
- **He's an erudite:** the weird-facts menu at the end is his reward currency — let him pick, and let him add facts he knows.
- **Don't be preachy:** Q9 (is $100 billion worth it?) is HIS opinion — no steering. Facts speak.
- **Ride the tangent:** space food, toilets, "can you see the Great Wall from space" (you can't, easily — good tangent) — 30–60 seconds, then "Good thought, Flight Director — log it and back to the mission."
- **China connection is built in** (Tiangong in Activities III and VII) — personal and real, use it.
- **The Russian module names are YOUR moment:** Zvezda, Zarya, Nauka, Prichal — say them with real pronunciation; for Mark it's proof the station really is half-Russian, and for you it's a rare native-speaker flex.
- **The 10-statement recap is a sprint:** keep it snappy, hands up/down, momentum over accuracy. It wakes him up, it doesn't test him.

Materials

- Screen-share ready: ISS live camera (I), ESA tour / ISS Explorer (II), Hadfield teeth OR sleep video (II), Heavens-Above (III), Tears in Space video (IV), PhET Gravity and Orbits (V), Spectrum pages 116–117 (VI), stations video + 4 station pictures (VII)
- The three reference screenshots in `screenshots/` (sky-map info panel, hatch labels, module map) — open them in a viewer before class; they are the visual aids for Activities II and III
- Annotation/whiteboard for the number hunt in Activity VI
- Optional: pre-checked flyover screenshot as backup for Activity III

Custom HTML needed

None. Per Alex's call: existing tools only (PhET, ESA tour / ISS Explorer, Heavens-Above, live feed, videos). The review quiz for the arc lives at the end of Part 2 verbally (Activity VIII) — if the arc later gets a Part 3 or you want a printed/HTML quiz after all, say the word and I'll build one in the established style.